Other Outcomes: Ordinal, Nominal, Count, Duration

Francisco Villamil

Applied Quantitative Methods II

MA in Social Sciences, Spring 2026

Today's goals

- Understand the GLM framework as a unifying structure
- Model ordinal outcomes with ordered logit
- Model nominal (multi-category) outcomes with multinomial logit
- Model count data with Poisson and negative binomial regression
- Model time-to-event data with survival analysis and the Cox model
- Know which model to reach for and why

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor	Link function	Distribution
$\eta_i = \mathbf{x}_i \boldsymbol{\beta}$	$g(\mu_i) = \eta_i$	$Y_i \sim F(\mu_i)$
A weighted sum of covariates	Connects mean to predictor	Data-generating process

- OLS is a GLM: identity link + normal distribution

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor	Link function	Distribution
$\eta_i = \mathbf{x}_i \boldsymbol{\beta}$	$g(\mu_i) = \eta_i$	$Y_i \sim F(\mu_i)$
A weighted sum of covariates	Connects mean to predictor	Data-generating process

- OLS is a GLM: identity link + normal distribution
- Logit is a GLM: logit link + Bernoulli distribution

One framework, many models

Generalized Linear Models (GLMs) share three components:

Linear predictor	Link function	Distribution
$\eta_i = \mathbf{x}_i \boldsymbol{\beta}$	$g(\mu_i) = \eta_i$	$Y_i \sim F(\mu_i)$
A weighted sum of covariates	Connects mean to predictor	Data-generating process

- OLS is a GLM: identity link + normal distribution
- Logit is a GLM: logit link + Bernoulli distribution
- Choosing a model = choosing a link + distribution

Matching outcome type to model

Outcome type	**Distribution**	**Link**	**Model / Function**
Continuous	Normal	Identity	OLS (`lm()`)
Binary (0/1)	Bernoulli	Logit	Logit (`glm(...,binomial)`)
Ordinal	Prop. odds	Cumul. logit	Ordered logit (`polr()`)
Nominal (>2 cat.)	Multinomial	Log-odds	Multinom. logit (`multinom()`)
Count (≥ 0)	Poisson	Log	Poisson (`glm(...,poisson)`)
Count (overdispersed)	Neg. Binomial	Log	NegBin (`glm.nb()`)
Duration (time)	—	—	Cox PH (`coxph()`)

First question: what is the data-generating process for your outcome?

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Ordinal outcomes:

When categories have an order but not equal spacing

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
  - → Survey: "How satisfied are you?" (1 = not at all, 5 = very)

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
  - → Survey: "How satisfied are you?" (1 = not at all, 5 = very)
  - → Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
  - → Survey: "How satisfied are you?" (1 = not at all, 5 = very)
  - → Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
  - → Democracy rating: Autocracy → Hybrid → Democracy

Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
  - → Survey: "How satisfied are you?" (1 = not at all, 5 = very)
  - → Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
  - → Democracy rating: Autocracy → Hybrid → Democracy
  - → Conflict intensity: No conflict → Low → Medium → High

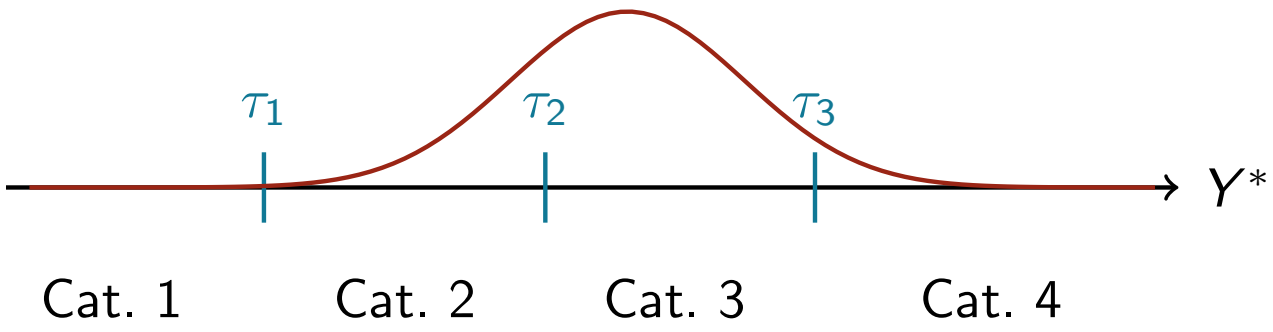
Ordinal outcomes are ordered categories

- Categories have a natural **ordering**, but not equal intervals
- Examples:
  - → Survey: "How satisfied are you?" (1 = not at all, 5 = very)
  - → Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
  - → Democracy rating: Autocracy → Hybrid → Democracy
  - → Conflict intensity: No conflict → Low → Medium → High
- The distance between "1" and "2" need not equal the distance between "2" and "3"

Ordinal outcomes are ordered categories

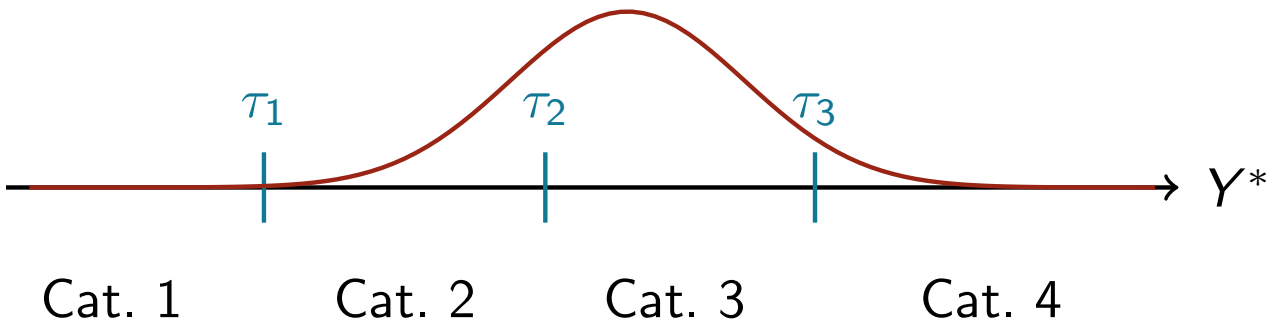
- Categories have a natural **ordering**, but not equal intervals
- Examples:
  - → Survey: "How satisfied are you?" (1 = not at all, 5 = very)
  - → Party ID: Strong Dem → Weak Dem → Ind → Weak Rep → Strong Rep
  - → Democracy rating: Autocracy → Hybrid → Democracy
  - → Conflict intensity: No conflict → Low → Medium → High
- The distance between "1" and "2" need not equal the distance between "2" and "3"
- OLS treats them as equally spaced — often wrong

The latent variable idea



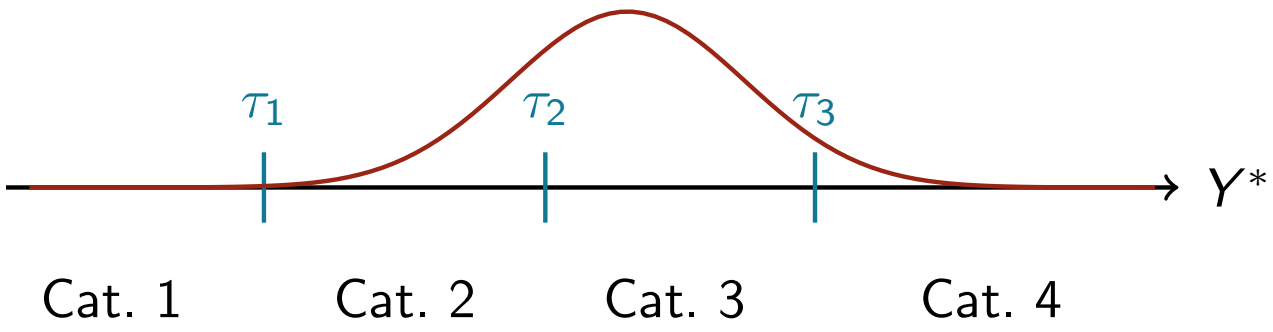
- Y^*: unobserved continuous variable (e.g., "true" satisfaction)

The latent variable idea



- Y^*: unobserved continuous variable (e.g., "true" satisfaction)
- We observe category j when $\tau_{j-1} < Y^* \leq \tau_j$

The latent variable idea



- Y^*: unobserved continuous variable (e.g., "true" satisfaction)
- We observe category j when $\tau_{j-1} < Y^* \leq \tau_j$
- Thresholds τ_j are **estimated** from the data

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i \boldsymbol{\beta}))}$$

- One set of $\boldsymbol{\beta}$ for all thresholds (**proportional odds assumption**)

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i \boldsymbol{\beta}))}$$

- One set of $\boldsymbol{\beta}$ for all thresholds (**proportional odds assumption**)
- $\beta_k > 0$ shifts Y^* upward $\rightarrow$ **higher** categories become more likely

Ordered logit: the model

Cumulative link model (proportional odds)

$$\Pr(Y_i \leq j \mid \mathbf{x}_i) = \frac{1}{1 + \exp(-(\tau_j - \mathbf{x}_i \boldsymbol{\beta}))}$$

- One set of $\boldsymbol{\beta}$ for all thresholds (**proportional odds assumption**)
- $\beta_k > 0$ shifts Y^* upward $\rightarrow$ **higher** categories become more likely
- $\Pr(Y_i = j) = \Pr(Y_i \leq j) - \Pr(Y_i \leq j - 1)$: probability for each category by differencing cumulative probabilities

Ordered logit in R

```
library(MASS); library(marginaleffects)
m_ord = polr(factor(satisfaction) ~ age + educ,
     data = mydata, Hess = TRUE)
avg_slopes(m_ord) # AMEs per predictor per category
```

- `Hess = TRUE`: stores the Hessian — required for SEs

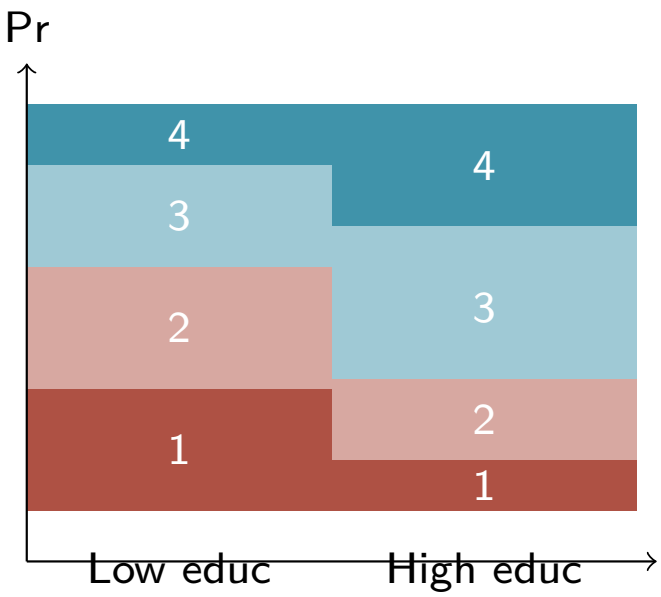
Ordered logit in R

```
library(MASS); library(marginaleffects)
m_ord = polr(factor(satisfaction) ~ age + educ,
     data = mydata, Hess = TRUE)
avg_slopes(m_ord) # AMEs per predictor per category
```

- `Hess = TRUE`: stores the Hessian — required for SEs
- Sign warning: raw `polr()` coefficients use a reversed sign convention — always interpret via `avg_slopes()`

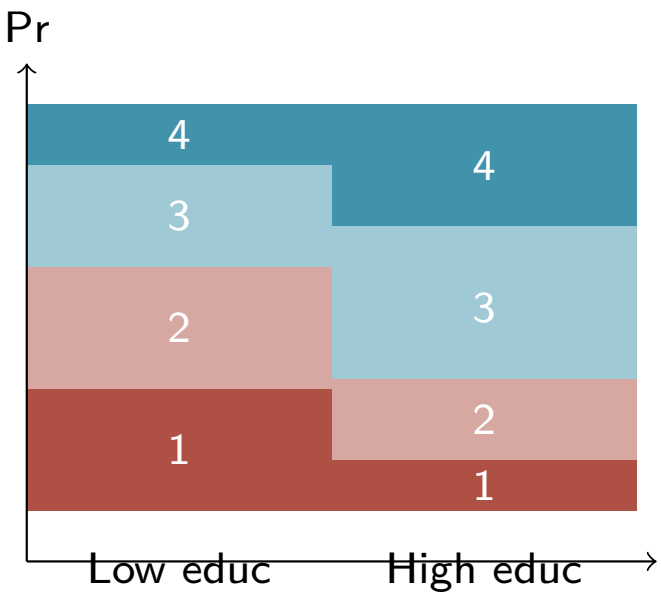
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive



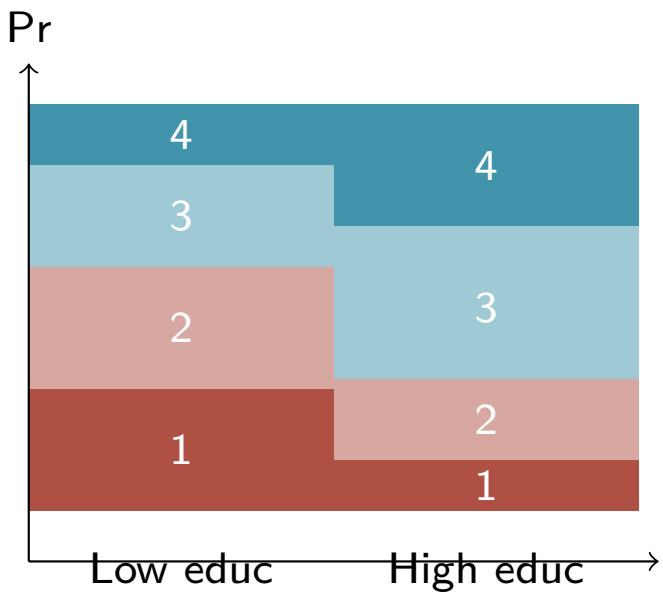
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects



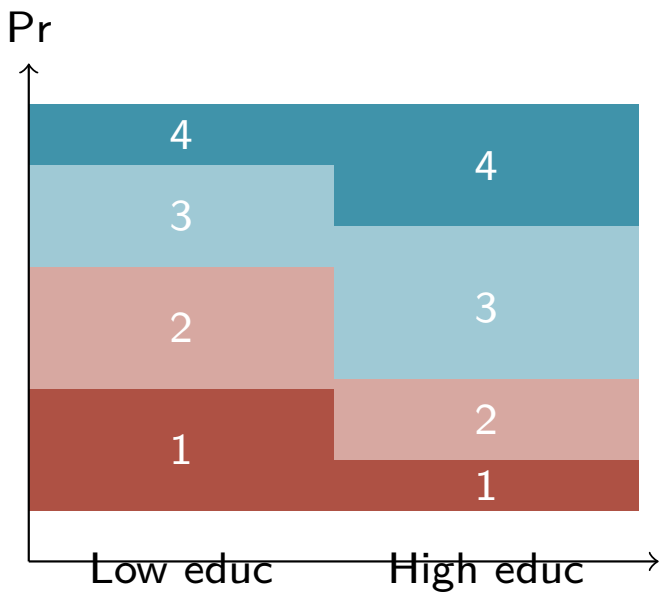
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects
  - → One AME per predictor *per category*



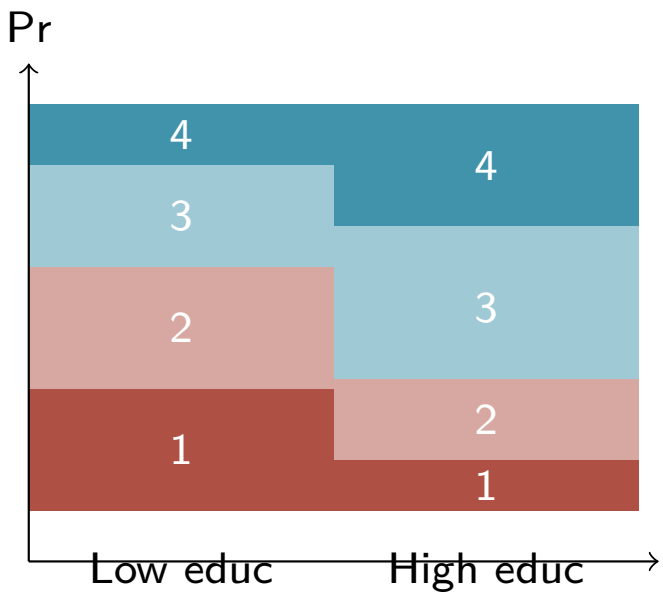
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects
  - → One AME per predictor *per category*
  - → "A one-unit increase in education changes the probability of category j by ΔP_j"



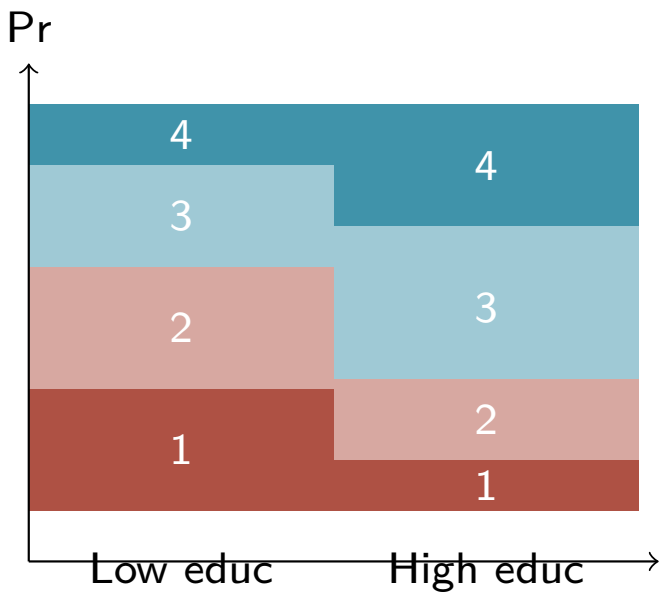
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects
  - → One AME per predictor *per category*
  - → "A one-unit increase in education changes the probability of category j by ΔP_j"
- **Best**: predicted probability plots



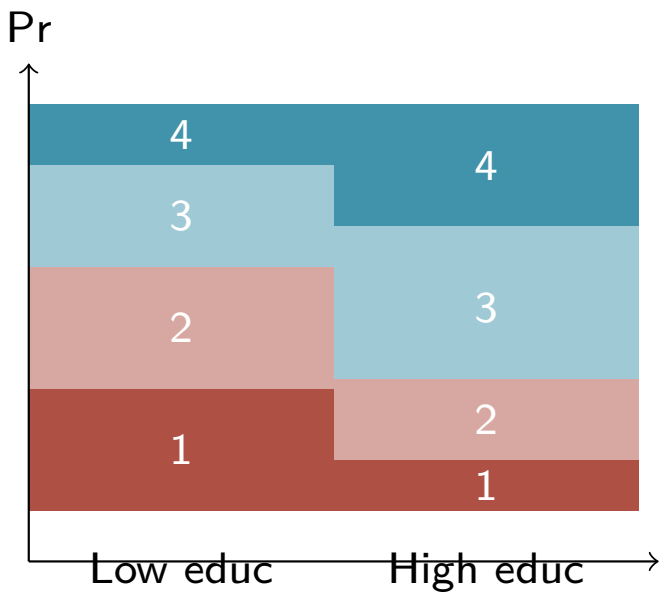
Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects
  - → One AME per predictor *per category*
  - → "A one-unit increase in education changes the probability of category j by ΔP_j"
- **Best**: predicted probability plots
  - → How does $\Pr(Y = j)$ change across X?



Interpreting ordered logit

- Raw coefficients: log-odds — not intuitive
- **Better**: average marginal effects
  - → One AME per predictor *per category*
  - → "A one-unit increase in education changes the probability of category j by ΔP_j"
- **Best**: predicted probability plots
  - → How does $\Pr(Y = j)$ change across X?
  - → Use `plot_predictions(m_ord, condition = "educ")`



Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Nominal outcomes:

Multiple unordered categories — vote choice, transport mode, conflict type

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:
  - → Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, . . .)

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:
  - → Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, . . .)
  - → Mode of transport (car, bus, train, bike, walk)

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:
  - → Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, . . .)
  - → Mode of transport (car, bus, train, bike, walk)
  - → Type of conflict (interstate, civil war, non-state)

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:
  - → Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, . . .)
  - → Mode of transport (car, bus, train, bike, walk)
  - → Type of conflict (interstate, civil war, non-state)
  - → Occupation category (manager, professional, manual, service)

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:
  - → Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, . . .)
  - → Mode of transport (car, bus, train, bike, walk)
  - → Type of conflict (interstate, civil war, non-state)
  - → Occupation category (manager, professional, manual, service)
- Ordered logit is wrong here — it would impose an ordering

When categories have no natural order

- **Nominal outcomes**: categories, but no ranking
- Examples:
  - → Vote choice in a multi-party election (PSOE, PP, Vox, Sumar, . . .)
  - → Mode of transport (car, bus, train, bike, walk)
  - → Type of conflict (interstate, civil war, non-state)
  - → Occupation category (manager, professional, manual, service)
- Ordered logit is wrong here — it would impose an ordering
- We need a model that treats all categories symmetrically

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\boldsymbol{\beta}_j' \mathbf{x}_i)}{\sum_{k=1}^{J} \exp(\boldsymbol{\beta}_k' \mathbf{x}_i)}$$

- J categories $\rightarrow$ $J-1$ sets of coefficients (one category is the **reference**)

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\boldsymbol{\beta}_j' \mathbf{x}_i)}{\sum_{k=1}^{J} \exp(\boldsymbol{\beta}_k' \mathbf{x}_i)}$$

- J categories $\rightarrow$ $J-1$ sets of coefficients (one category is the **reference**)
- Reference category: $\boldsymbol{\beta}_1 = \mathbf{0}$ by convention

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\boldsymbol{\beta}_j' \mathbf{x}_i)}{\sum_{k=1}^{J} \exp(\boldsymbol{\beta}_k' \mathbf{x}_i)}$$

- J categories $\rightarrow$ $J-1$ sets of coefficients (one category is the **reference**)
- Reference category: $\boldsymbol{\beta}_1 = \mathbf{0}$ by convention
- Each β_{jk}: effect of x_k on log-odds of category j **vs. reference**

Multinomial logit: the model

Softmax probabilities

$$\Pr(Y_i = j \mid \mathbf{x}_i) = \frac{\exp(\boldsymbol{\beta}_j' \mathbf{x}_i)}{\sum_{k=1}^{J} \exp(\boldsymbol{\beta}_k' \mathbf{x}_i)}$$

- J categories $\rightarrow$ $J-1$ sets of coefficients (one category is the **reference**)
- Reference category: $\boldsymbol{\beta}_1 = \mathbf{0}$ by convention
- Each β_{jk}: effect of x_k on log-odds of category j **vs. reference**
- All predicted probabilities sum to 1: $\sum_{j=1}^{J} \Pr(Y_i = j) = 1$

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
      data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
```

- First category = reference; coefficients = log-odds vs. reference

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
      data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
```

- First category = reference; coefficients = log-odds vs. reference
- Change reference: `relevel(factor(vote), ref = "Labour")`

Multinomial logit in R

```
library(nnet)

m_mnom = multinom(vote ~ age + ideology + income,
      data = beps)

# AMEs (one per predictor per category)
avg_slopes(m_mnom)
```

- First category = reference; coefficients = log-odds vs. reference
- Change reference: `relevel(factor(vote), ref = "Labour")`
- Interpret via `avg_slopes()` or `plot_predictions()`

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp\big((\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)'\mathbf{x}\big)$$

- The famous **red bus / blue bus** problem:

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp\big((\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)'\mathbf{x}\big)$$

- The famous **red bus / blue bus** problem:
  - → 50% car, 50% red bus — now introduce a blue bus

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp\big((\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)'\mathbf{x}\big)$$

- The famous **red bus / blue bus** problem:
  - → 50% car, 50% red bus — now introduce a blue bus
  - → IIA implies: 33% car, 33% red bus, 33% blue bus

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp\big((\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)'\mathbf{x}\big)$$

- The famous **red bus / blue bus** problem:
  - → 50% car, 50% red bus — now introduce a blue bus
  - → IIA implies: 33% car, 33% red bus, 33% blue bus
  - → Reality: 50% car, 25% red bus, 25% blue bus

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp\big((\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)'\mathbf{x}\big)$$

- The famous **red bus / blue bus** problem:
  - → 50% car, 50% red bus — now introduce a blue bus
  - → IIA implies: 33% car, 33% red bus, 33% blue bus
  - → Reality: 50% car, 25% red bus, 25% blue bus
- IIA holds when alternatives are **genuinely distinct**

The IIA assumption

Independence of Irrelevant Alternatives (IIA): the odds ratio between any two alternatives is **unaffected** by the presence or absence of other alternatives.

$$\frac{\Pr(Y = j)}{\Pr(Y = k)} = \exp\big((\boldsymbol{\beta}_j - \boldsymbol{\beta}_k)'\mathbf{x}\big)$$

- The famous **red bus / blue bus** problem:
  - → 50% car, 50% red bus — now introduce a blue bus
  - → IIA implies: 33% car, 33% red bus, 33% blue bus
  - → Reality: 50% car, 25% red bus, 25% blue bus
- IIA holds when alternatives are **genuinely distinct**

You are modeling vote choice in a Spanish election with five parties.

Two parties are both center-left. Should you worry about the IIA assumption?

What would you do?

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

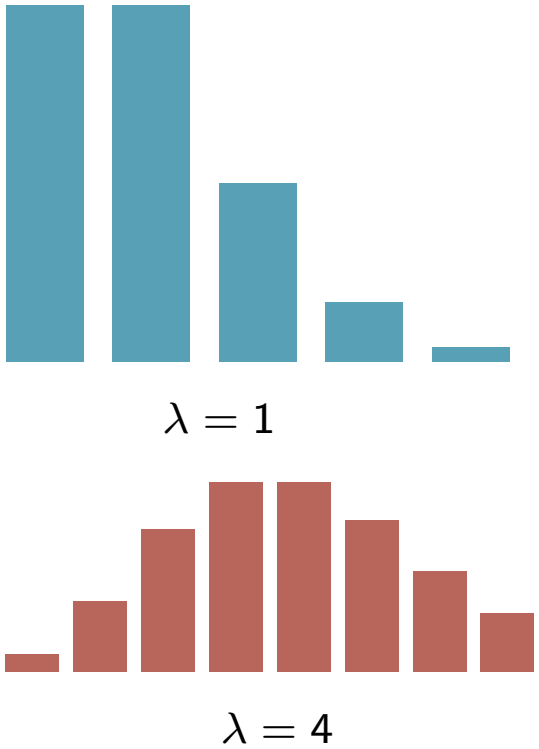
Model Selection in Practice

Count data:

Non-negative integers — conflicts, protests, bills, articles

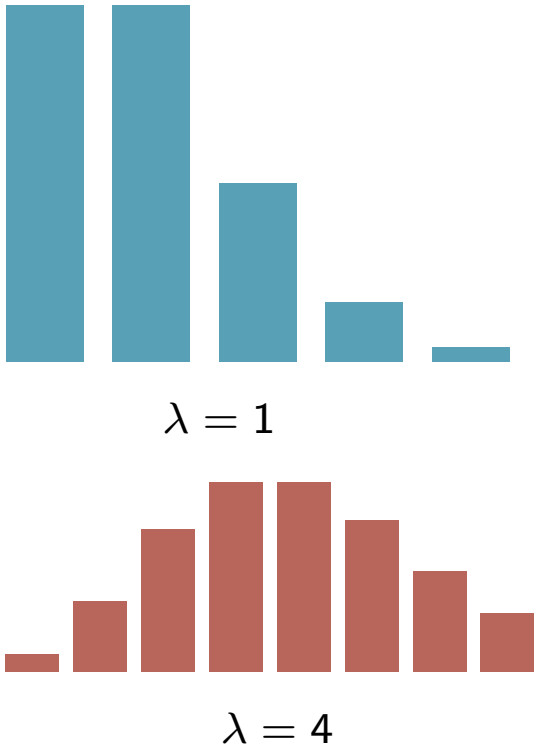
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...



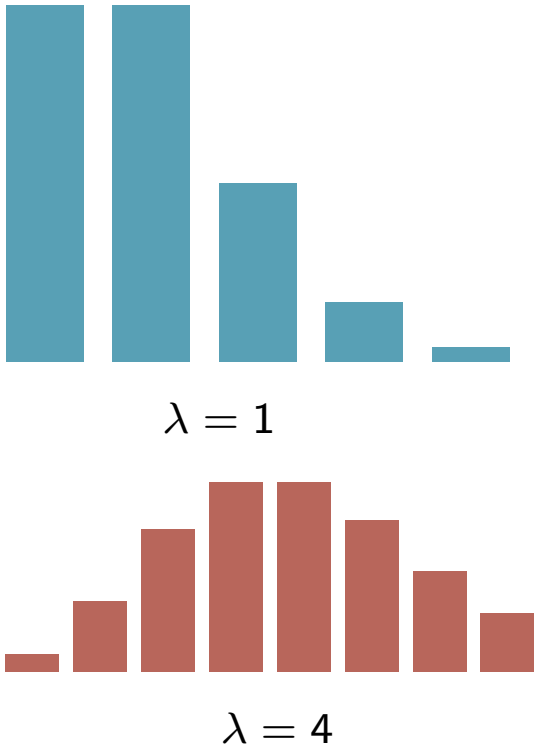
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks



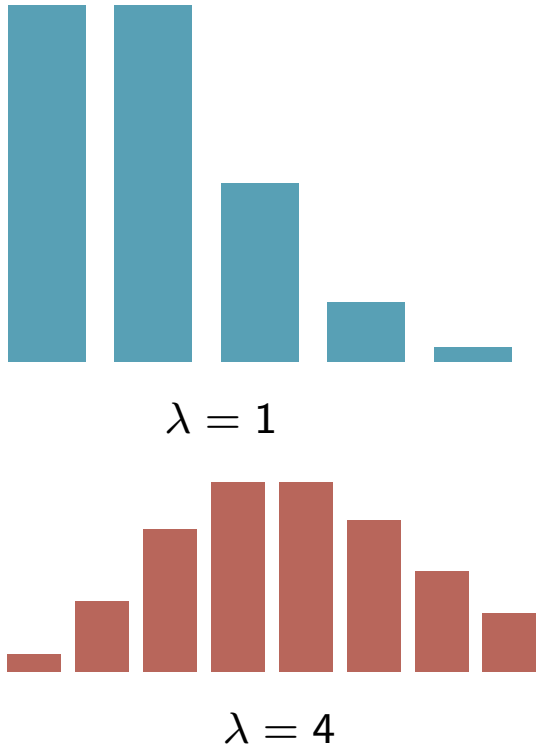
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**



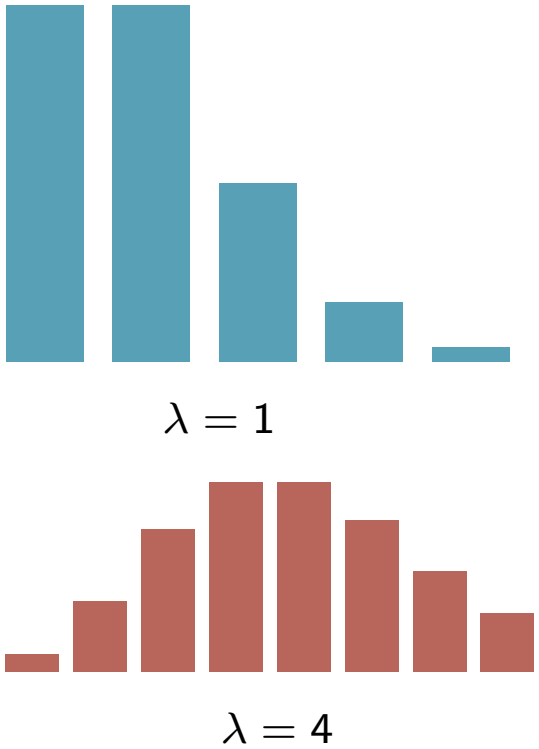
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**
  - → Counts ≥ 0: OLS can predict negative values



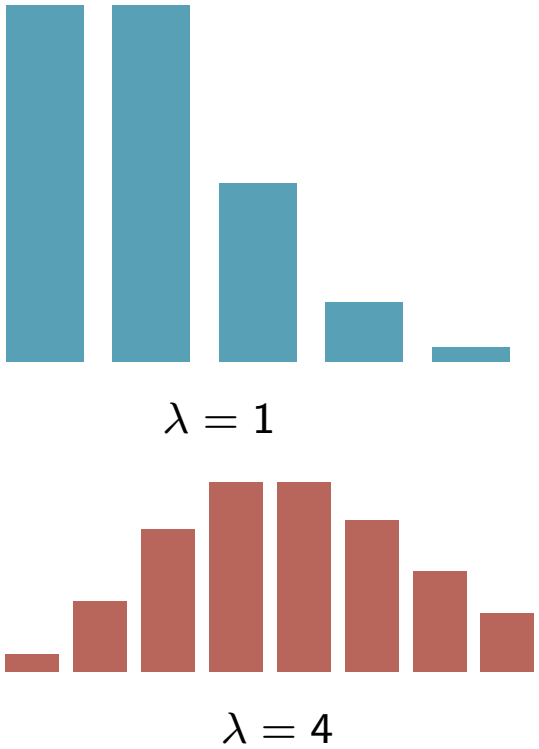
Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**
  - → Counts ≥ 0: OLS can predict negative values
  - → Variance increases with the mean



Count outcomes

- Outcomes that are non-negative integers: 0, 1, 2, 3, ...
- Examples: conflict events, bills, protests, publications, attacks
- **Why not OLS?**
  - → Counts ≥ 0: OLS can predict negative values
  - → Variance increases with the mean
  - → Strongly right-skewed; often many zeros



The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i \boldsymbol{\beta})$$

$$\Leftrightarrow \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i \boldsymbol{\beta})$$

$$\Leftrightarrow \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i \boldsymbol{\beta})$$

$$\Leftrightarrow \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative
- Coefficient interpretation: $\exp(\beta_k) =$ **multiplicative** change in expected count

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i \boldsymbol{\beta})$$

$$\Leftrightarrow \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative
- Coefficient interpretation: $\exp(\beta_k) =$ **multiplicative** change in expected count
  - $\rightarrow$ $\beta_k = 0.3 \Rightarrow \exp(0.3) \approx 1.35$: 35% increase in expected count

The Poisson model

Poisson regression

$$Y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \exp(\mathbf{x}_i \boldsymbol{\beta})$$

$$\Leftrightarrow \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

- $\mu_i = E[Y_i \mid \mathbf{x}_i]$: expected count (always > 0)
- The **log link** ensures predictions are always non-negative
- Coefficient interpretation: $\exp(\beta_k) =$ **multiplicative** change in expected count
  - $\rightarrow$ $\beta_k = 0.3 \Rightarrow \exp(0.3) \approx 1.35$: 35% increase in expected count
- Poisson assumption: $\text{Var}(Y_i) = E[Y_i]$ (mean = variance)

Poisson in R and interpretation

```
# Fit Poisson model
m_pois = glm(n_conflicts ~ gdp_pc + pop_l,
     family = "poisson", data = mydata)

summary(m_pois)
exp(coef(m_pois)) # incidence rate ratios
```

Raw coefficient

$\hat{\beta}_{\texttt{gdp_pc}} = -0.24$

Hard to interpret directly

Exponentiated

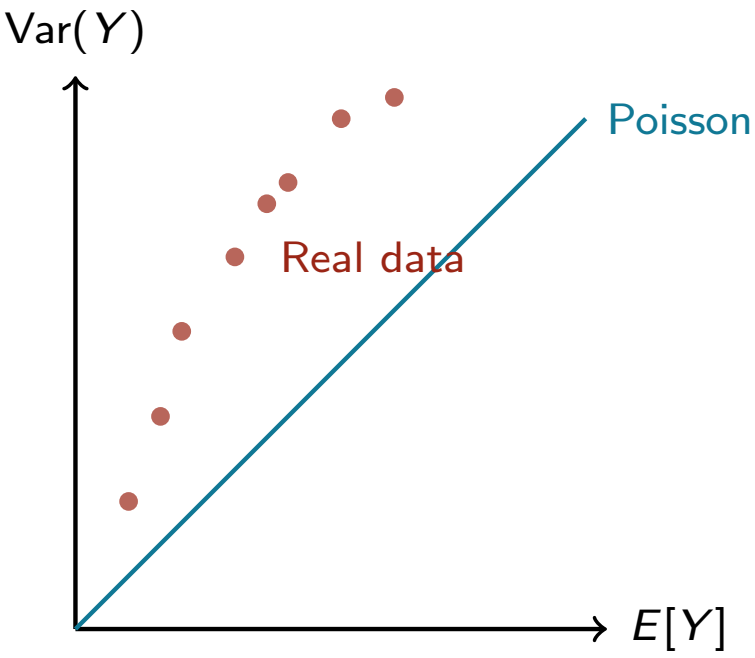
$\exp(-0.24) \approx 0.79$

$\rightarrow$ 21% fewer conflicts per unit GDP

Also: `avg_slopes(m_pois)` gives AMEs on count scale

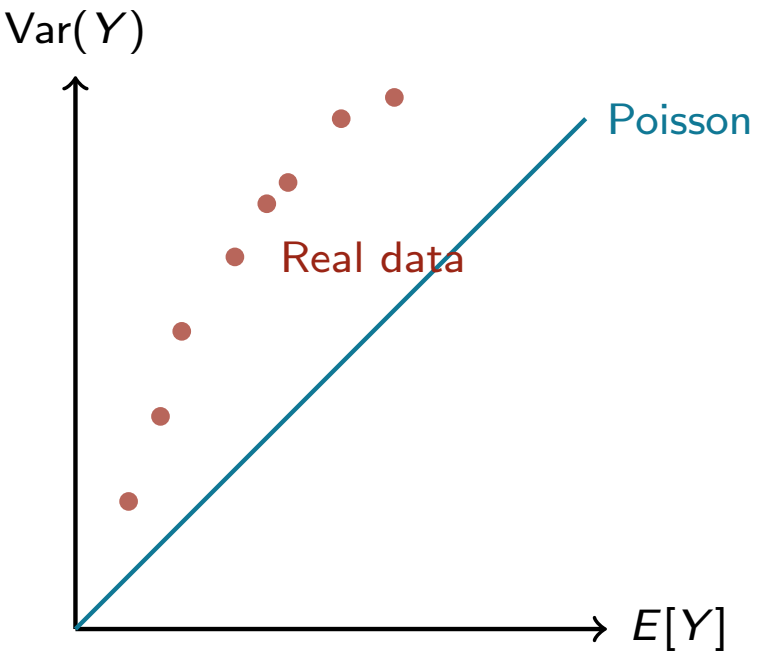
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$



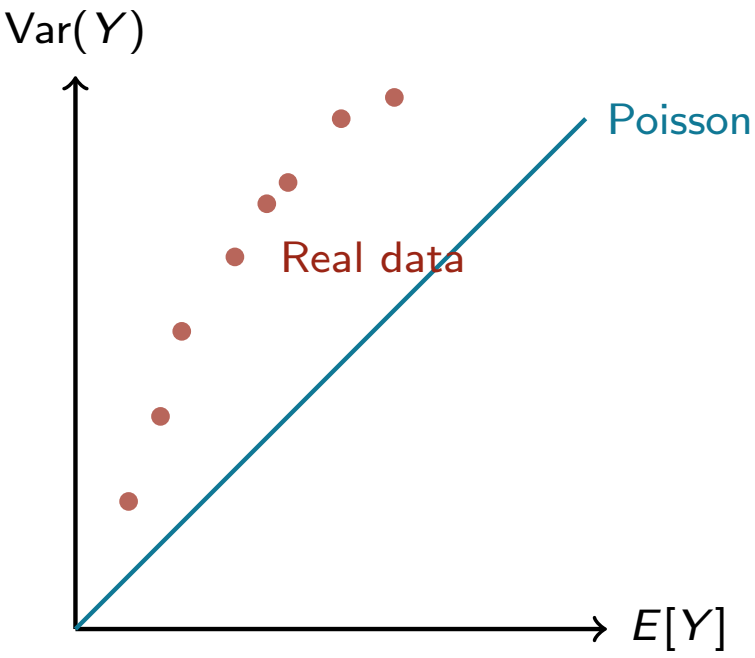
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always: $\text{Var}(Y) \gg E[Y]$



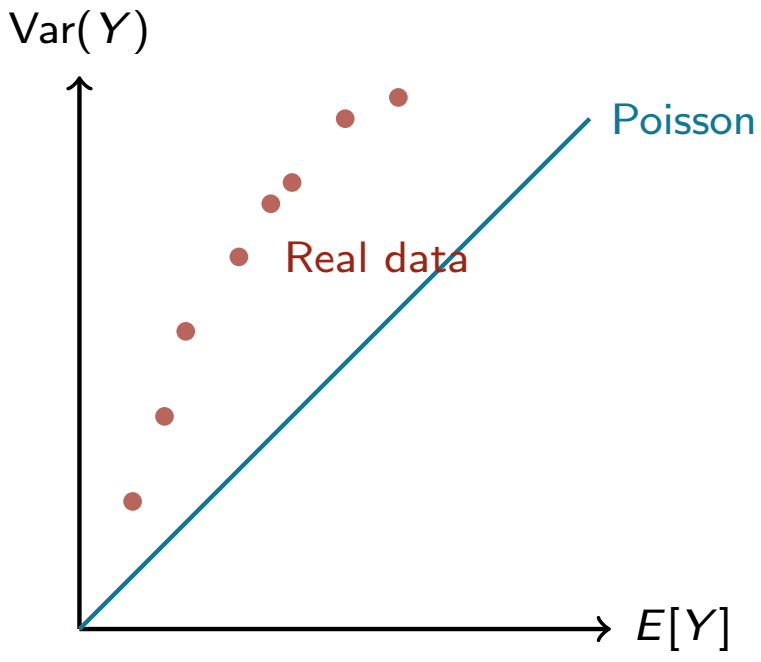
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always: $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:



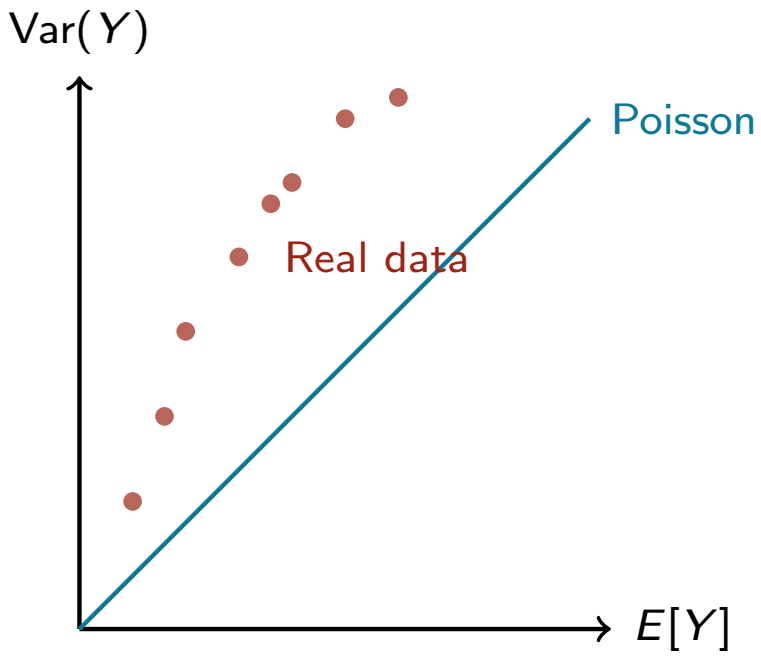
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always: $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
  - $\rightarrow$ Standard errors too **small**



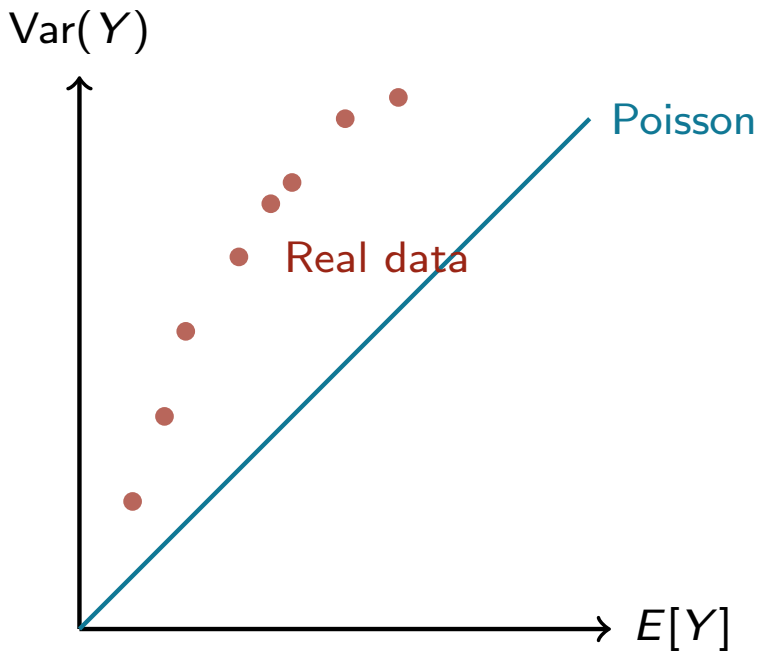
The overdispersion problem

- Poisson requires: $\text{Var}(Y) = E[Y]$
- Real data almost always: $\text{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
  - $\rightarrow$ Standard errors too **small**
  - $\rightarrow$ p-values too **small**



The overdispersion problem

- Poisson requires: $\mathrm{Var}(Y) = E[Y]$
- Real data almost always: $\mathrm{Var}(Y) \gg E[Y]$
- Consequences of ignoring overdispersion:
  - $\rightarrow$ Standard errors too **small**
  - $\rightarrow$ p-values too **small**
  - $\rightarrow$ False positives



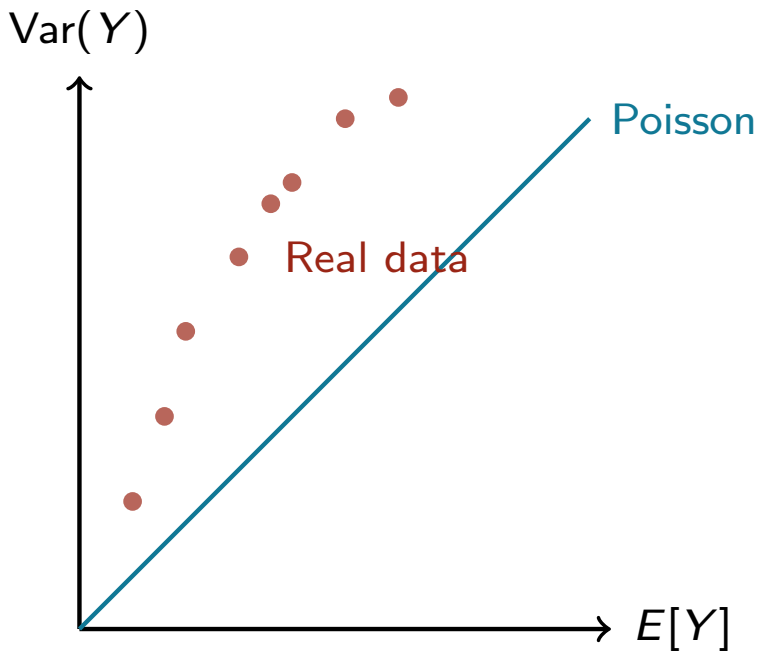
The overdispersion problem

- Poisson requires: $\mathrm{Var}(Y) = E[Y]$

- Real data almost always:
  $\mathrm{Var}(Y) \gg E[Y]$

- Consequences of ignoring overdispersion:
  - $\rightarrow$ Standard errors too **small**
  - $\rightarrow$ p-values too **small**
  - $\rightarrow$ False positives

- Quick check:
  Residual deviance $\gg$ df



Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
  - $\rightarrow$ $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
  - $\rightarrow$ $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)
  - $\rightarrow$ Small θ: high overdispersion

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
  - $\rightarrow$ $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)
  - $\rightarrow$ Small θ: high overdispersion
- Same interpretation as Poisson: $\exp(\beta_k)$ = incidence rate ratio

Negative binomial regression

Negative Binomial model

$$Y_i \sim \text{NegBin}(\mu_i, \theta), \quad \log(\mu_i) = \mathbf{x}_i \boldsymbol{\beta}$$

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\theta}$$

- $\theta > 0$: dispersion parameter estimated from the data
  - $\rightarrow$ $\theta \rightarrow \infty$: approaches Poisson (no overdispersion)
  - $\rightarrow$ Small θ: high overdispersion
- Same interpretation as Poisson: $\exp(\beta_k)$ = incidence rate ratio
- In R: `MASS::glm.nb(y ~ x1 + x2)`

Count models: a summary

Model	R function	When to use
Poisson	`glm(...,poisson)`	Equidispersed counts
Negative binomial	`MASS::glm.nb()`	Overdispersed counts
Zero-inflated Poisson	`pscl::zeroinfl()`	Excess zeros, equidisp.
Zero-inflated NegBin	`pscl::zeroinfl()`	Excess zeros, overdisp.

- **Default**: start Poisson, test overdispersion, switch to NegBin if needed

Count models: a summary

Model	R function	When to use
Poisson	`glm(...,poisson)`	Equidispersed counts
Negative binomial	`MASS::glm.nb()`	Overdispersed counts
Zero-inflated Poisson	`pscl::zeroinfl()`	Excess zeros, equidisp.
Zero-inflated NegBin	`pscl::zeroinfl()`	Excess zeros, overdisp.

- **Default**: start Poisson, test overdispersion, switch to NegBin if needed
- Zero-inflation: two-process model (structural vs. count zeros)

Count models: a summary

Model	R function	When to use
Poisson	`glm(...,poisson)`	Equidispersed counts
Negative binomial	`MASS::glm.nb()`	Overdispersed counts
Zero-inflated Poisson	`pscl::zeroinfl()`	Excess zeros, equidisp.
Zero-inflated NegBin	`pscl::zeroinfl()`	Excess zeros, overdisp.

- **Default**: start Poisson, test overdispersion, switch to NegBin if needed
- Zero-inflation: two-process model (structural vs. count zeros)
- All share the log link: $\exp(\hat{\beta})$ = incidence rate ratio (IRR)

Discussion

You model the number of conflict events per country-year.

After fitting a Poisson model, you check the output: `Residual deviance: 847 on 142 df.`

What does this tell you? What would you do next?

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Duration analysis:

Time until an event — war onset, cabinet survival, regime collapse

Time-to-event outcomes

- Outcomes that record **how long until something happens**

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office
  - → Time to re-arrest after prison release

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office
  - → Time to re-arrest after prison release
  - → How long an authoritarian regime survives

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office
  - → Time to re-arrest after prison release
  - → How long an authoritarian regime survives
  - → Time until a bill is passed (from introduction)

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office
  - → Time to re-arrest after prison release
  - → How long an authoritarian regime survives
  - → Time until a bill is passed (from introduction)
- Why not OLS on duration?

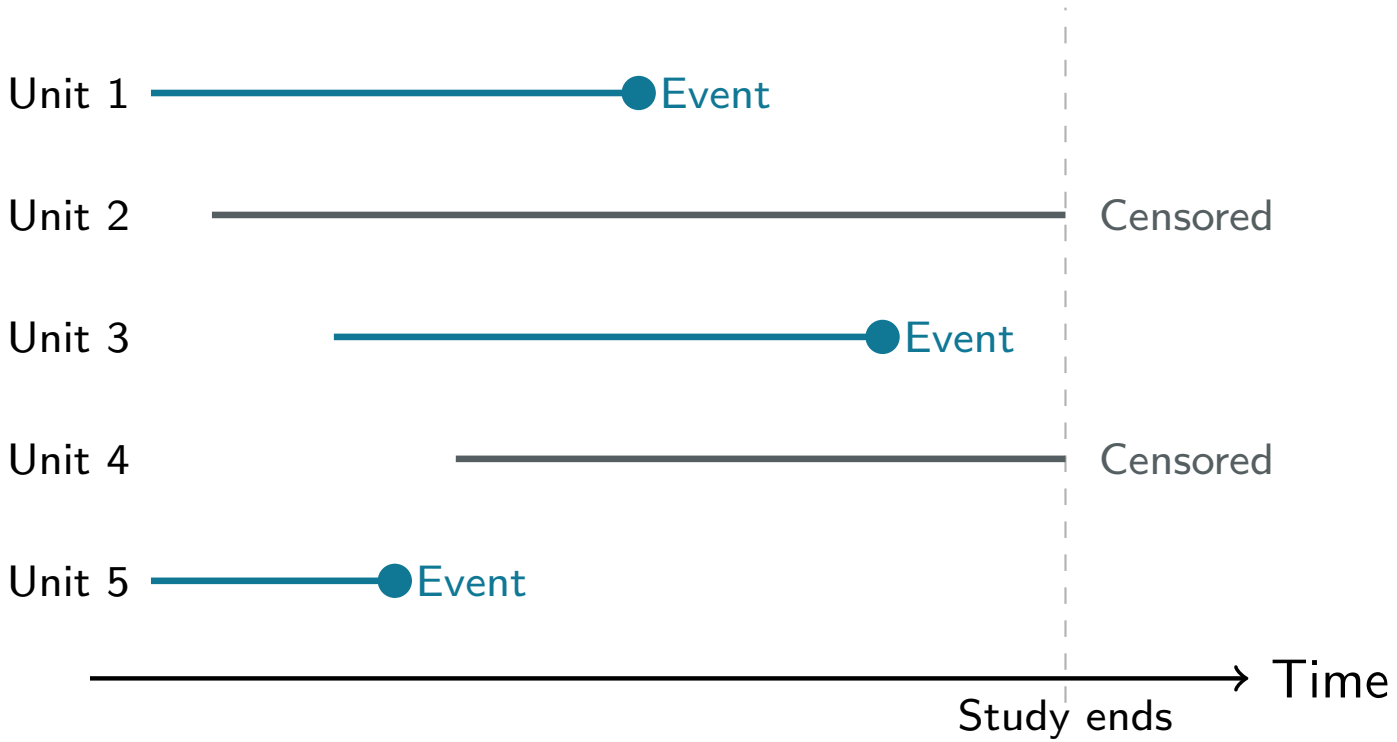
Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office
  - → Time to re-arrest after prison release
  - → How long an authoritarian regime survives
  - → Time until a bill is passed (from introduction)
- Why not OLS on duration?
  - → Durations are ≥ 0 and right-skewed

Time-to-event outcomes

- Outcomes that record **how long until something happens**
- Examples:
  - → Time until a war begins (from peace)
  - → Duration of a coalition government in office
  - → Time to re-arrest after prison release
  - → How long an authoritarian regime survives
  - → Time until a bill is passed (from introduction)
- Why not OLS on duration?
  - → Durations are ≥ 0 and right-skewed
  - → **Censoring**: many observations end without the event occurring

The censoring problem



Censored units: event did not occur during observation window

Discussion

A researcher studies how long democracies survive.

She drops all country-spells that are still ongoing at the end of the dataset, keeping only countries where a democratic breakdown was observed.

What is wrong with this decision?

Key concepts: survival and hazard

Survival function $S(t)$

$S(t) = \Pr(T > t)$

Probability of surviving past time t

$S(0) = 1$, $S(\infty) = 0$

Monotonically non-increasing

Hazard function $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t, conditional on survival to t

"Danger per unit time at time t"

- High $h(t)$: event is likely soon

Key concepts: survival and hazard

Survival function $S(t)$

$S(t) = \Pr(T > t)$

Probability of surviving past time t

$S(0) = 1$, $S(\infty) = 0$

Monotonically non-increasing

Hazard function $h(t)$

$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t, conditional on survival to t

"Danger per unit time at time t"

- High $h(t)$: event is likely soon
- $S(t)$ and $h(t)$ are two views of the same object

Key concepts: survival and hazard

Survival function $S(t)$

$S(t) = \Pr(T > t)$

Probability of surviving past time t

$S(0) = 1$, $S(\infty) = 0$

Monotonically non-increasing

Hazard function $h(t)$

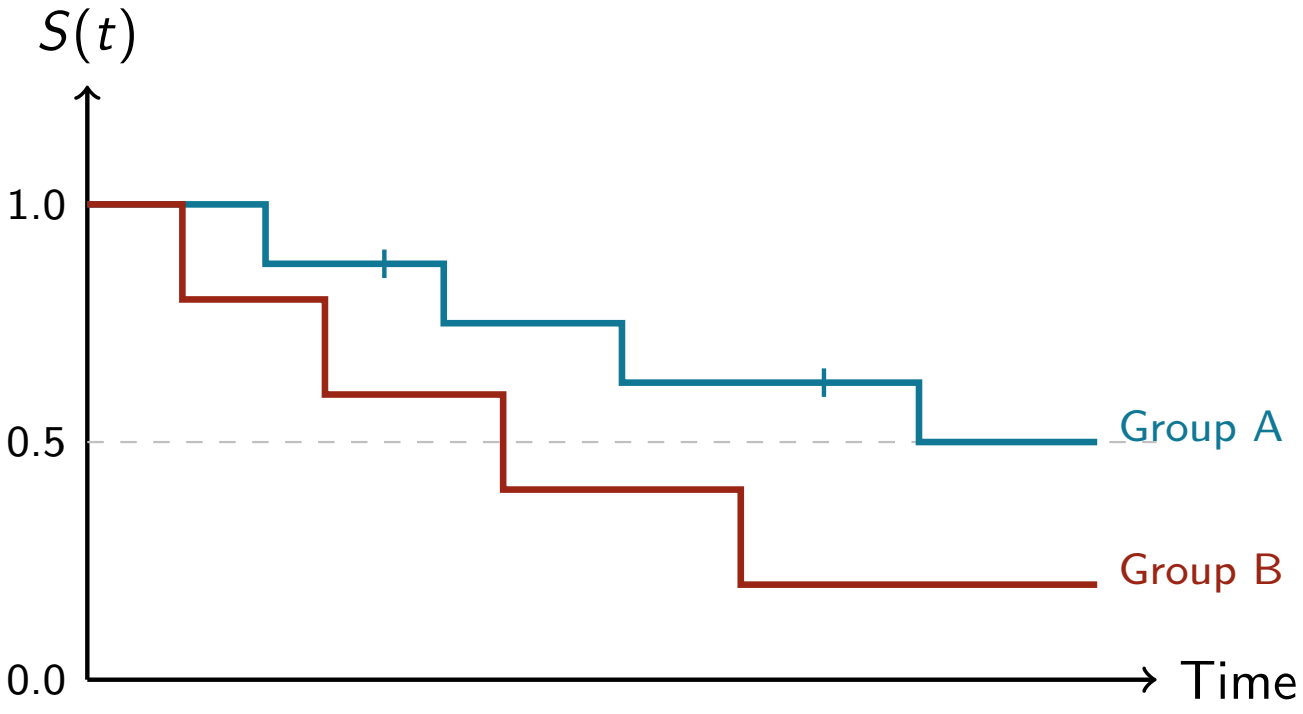
$$h(t) = \frac{f(t)}{S(t)}$$

Instantaneous rate of event at t, conditional on survival to t

"Danger per unit time at time t"

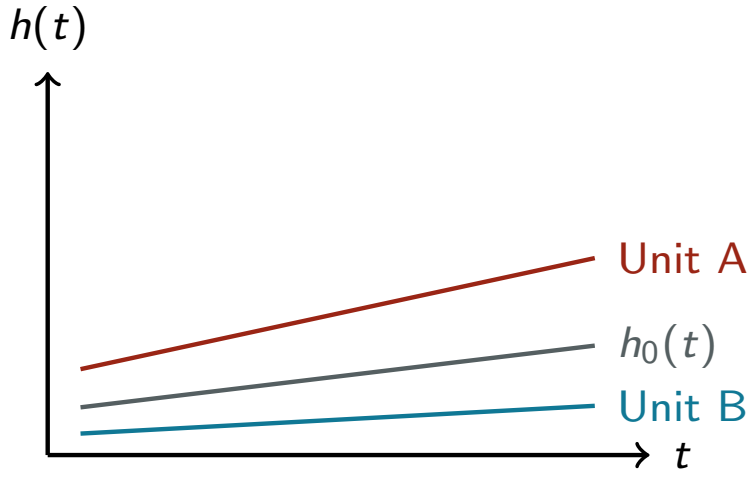
- High $h(t)$: event is likely soon
- $S(t)$ and $h(t)$ are two views of the same object
- Covariates shift $h(t)$ — that is what regression models

Kaplan-Meier: visualizing survival



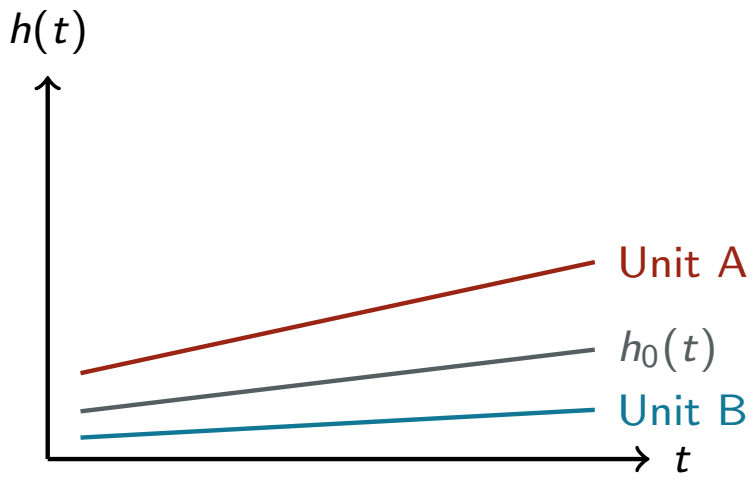
Kaplan-Meier estimator: non-parametric, no covariates, visual comparison of groups

The proportional hazards idea



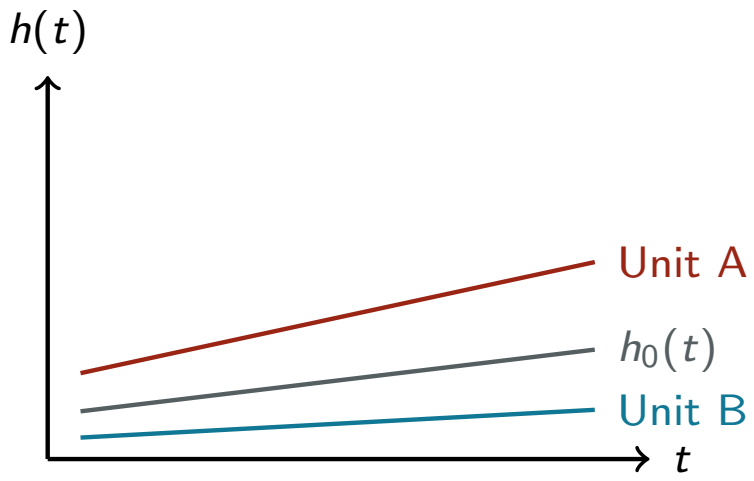
- **Key idea**: each unit has its own hazard, but all are proportional shifts of the same baseline $h_0(t)$

The proportional hazards idea



- **Key idea**: each unit has its own hazard, but all are proportional shifts of the same baseline $h_0(t)$
- The ratio $h_i(t)/h_j(t)$ is **constant over time** — curves never cross

The proportional hazards idea



- **Key idea**: each unit has its own hazard, but all are proportional shifts of the same baseline $h_0(t)$
- The ratio $h_i(t)/h_j(t)$ is **constant over time** — curves never cross
- Violation: if the gap between lines closes or reverses over time $\Rightarrow$ PH assumption fails

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i \boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i \boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i \boldsymbol{\beta})$: multiplicative shift of the baseline

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i \boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i \boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i \boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i \boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*
- $\exp(\beta_k)$: **hazard ratio** for a one-unit increase in x_k

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i \boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i \boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*
- $\exp(\beta_k)$: **hazard ratio** for a one-unit increase in x_k
  - $\rightarrow$ $\exp(\beta_k) > 1$: higher hazard (event happens sooner)

Cox proportional hazards model

Cox PH model

$$h_i(t) = h_0(t) \cdot \exp(\mathbf{x}_i \boldsymbol{\beta})$$

- $h_0(t)$: **baseline hazard** — left completely unspecified (semiparametric)
- $\exp(\mathbf{x}_i \boldsymbol{\beta})$: multiplicative shift of the baseline
- **Proportional hazards assumption**: the hazard ratio between any two units is *constant over time*
- $\exp(\beta_k)$: **hazard ratio** for a one-unit increase in x_k
  - $\rightarrow$ $\exp(\beta_k) > 1$: higher hazard (event happens sooner)
  - $\rightarrow$ $\exp(\beta_k) < 1$: lower hazard (event happens later)

Cox model in R

```
library(survival); library(survminer)

# Kaplan-Meier estimator
km_fit = survfit(Surv(time, event) ~ group, data = d)
ggsurvplot(km_fit, conf.int = TRUE)

# Cox PH model
cox_fit = coxph(Surv(time, event) ~ x1 + x2, data = d)
summary(cox_fit) # shows hazard ratios
exp(coef(cox_fit))
```

Interpreting the Cox model

	Coef	**Exp(coef)**	**SE**	***p*-value**
`gdp_pc`	-0.31	0.73	0.08	< 0.001
`military`	0.64	1.90	0.15	0.001
`log_pop`	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\texttt{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event

Interpreting the Cox model

	Coef	**Exp(coef)**	**SE**	***p*-value**
`gdp_pc`	-0.31	0.73	0.08	< 0.001
`military`	0.64	1.90	0.15	0.001
`log_pop`	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\texttt{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event
- $\exp(\hat{\beta}_{\texttt{military}}) = 1.90$: military regimes face 90% **higher** hazard

Interpreting the Cox model

	Coef	**Exp(coef)**	**SE**	***p*-value**
`gdp_pc`	-0.31	0.73	0.08	< 0.001
`military`	0.64	1.90	0.15	0.001
`log_pop`	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\texttt{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event
- $\exp(\hat{\beta}_{\texttt{military}}) = 1.90$: military regimes face 90% **higher** hazard
- Hazard ratio < 1: covariate *reduces* event rate $\rightarrow$ *increases* survival

Interpreting the Cox model

	Coef	**Exp(coef)**	**SE**	***p*-value**
`gdp_pc`	-0.31	0.73	0.08	< 0.001
`military`	0.64	1.90	0.15	0.001
`log_pop`	0.18	1.20	0.10	0.07

- $\exp(\hat{\beta}_{\texttt{gdp_pc}}) = 0.73$: richer countries have 27% **lower** hazard of event
- $\exp(\hat{\beta}_{\texttt{military}}) = 1.90$: military regimes face 90% **higher** hazard
- Hazard ratio < 1: covariate *reduces* event rate $\rightarrow$ *increases* survival
- Hazard ratio > 1: covariate *increases* event rate $\rightarrow$ *decreases* survival

Roadmap

The GLM Framework

Ordinal Outcomes

Nominal / Multinomial Outcomes

Count Data

Duration Analysis

Model Selection in Practice

Pulling it together:

Which model for which data?

Choosing the right model

If your outcome is...	**Use**	**Key check**
Continuous	OLS	Linearity, homoskedasticity
Binary (0/1)	Logit / LPM	Rare events $\rightarrow$ prefer logit
Ordered categories	Ordered logit	Test PO assumption (Brant)
Unordered categories	Multinomial logit	IIA plausible?
Non-negative integer	Poisson	Test overdispersion
Overdispersed count	Neg. Binomial	Often the safe default
Many zeros	Zero-inflated	Two-process data?
Time until event	Cox PH	Test PH assumption

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R
- **Step 4**: Check assumptions

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R
- **Step 4**: Check assumptions
  - $\rightarrow$ Counts: overdispersion test $\rightarrow$ switch to NegBin if needed

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R
- **Step 4**: Check assumptions
  - $\rightarrow$ Counts: overdispersion test $\rightarrow$ switch to NegBin if needed
  - $\rightarrow$ Ordinal: Brant test for proportional odds

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R
- **Step 4**: Check assumptions
  - → Counts: overdispersion test → switch to NegBin if needed
  - → Ordinal: Brant test for proportional odds
  - → Nominal: IIA plausibility (substantive reasoning)

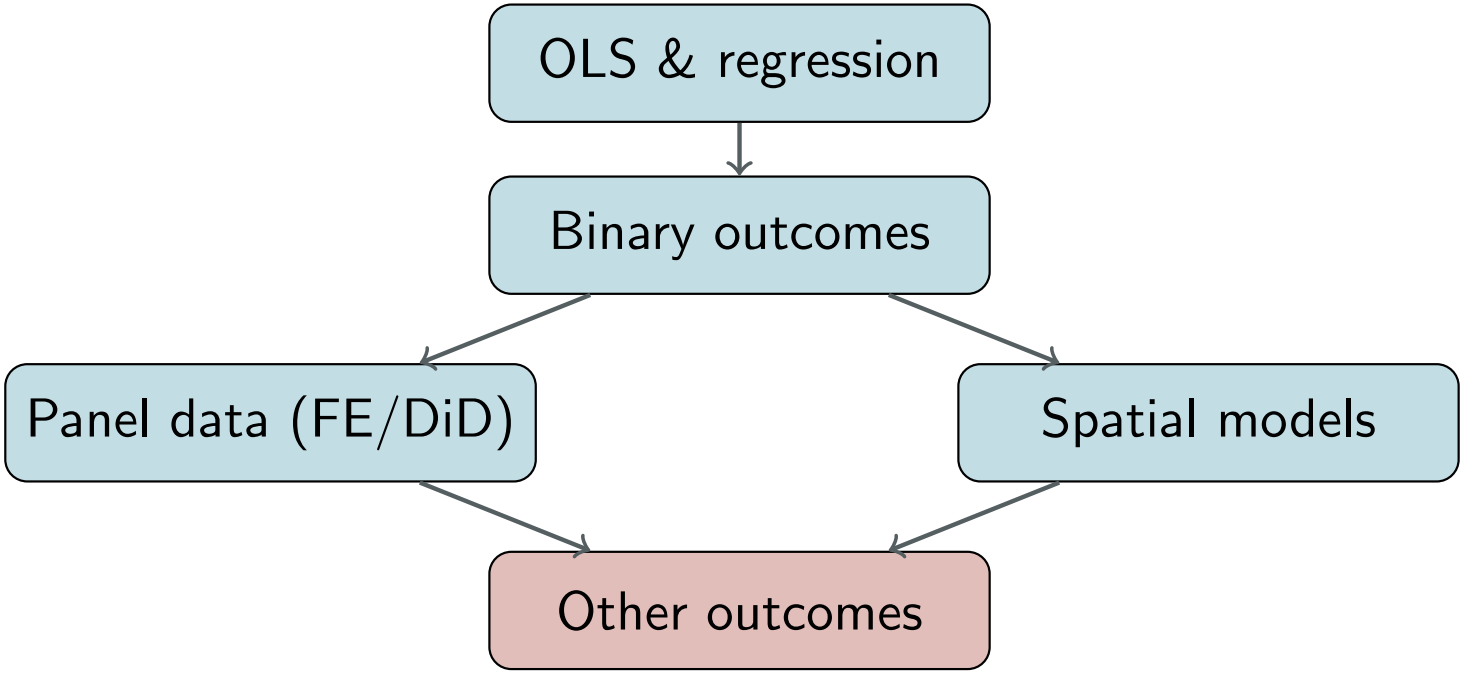
Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R
- **Step 4**: Check assumptions
  - → Counts: overdispersion test → switch to NegBin if needed
  - → Ordinal: Brant test for proportional odds
  - → Nominal: IIA plausibility (substantive reasoning)
  - → Duration: Schoenfeld residuals for PH assumption

Practical checklist

- **Step 1**: What is the DGP? Identify the outcome type
- **Step 2**: Choose model family accordingly (see table)
- **Step 3**: Estimate the model in R
- **Step 4**: Check assumptions
  - → Counts: overdispersion test → switch to NegBin if needed
  - → Ordinal: Brant test for proportional odds
  - → Nominal: IIA plausibility (substantive reasoning)
  - → Duration: Schoenfeld residuals for PH assumption
- **Steps 5–6**: Interpret via AMEs / predicted probabilities (**never raw coefficients alone**); visualize results

What you have learned this semester



Today: ordinal, nominal, count, duration

For the exam

- Be able to **identify** the right model given an outcome description

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)
  - → Equidispersion (Poisson)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)
  - → Equidispersion (Poisson)
  - → Proportional hazards (Cox)

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)
  - → Equidispersion (Poisson)
  - → Proportional hazards (Cox)
- Interpret coefficients correctly:

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)
  - → Equidispersion (Poisson)
  - → Proportional hazards (Cox)
- Interpret coefficients correctly:
  - → Ordered/multinomial logit: AMEs or predicted probabilities

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)
  - → Equidispersion (Poisson)
  - → Proportional hazards (Cox)
- Interpret coefficients correctly:
  - → Ordered/multinomial logit: AMEs or predicted probabilities
  - → Poisson/NegBin: $\exp(\hat{\beta})$ = incidence rate ratio

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - $\rightarrow$ Proportional odds (ordered logit)
  - $\rightarrow$ IIA (multinomial logit)
  - $\rightarrow$ Equidispersion (Poisson)
  - $\rightarrow$ Proportional hazards (Cox)
- Interpret coefficients correctly:
  - $\rightarrow$ Ordered/multinomial logit: AMEs or predicted probabilities
  - $\rightarrow$ Poisson/NegBin: $\exp(\hat{\beta})$ = incidence rate ratio
  - $\rightarrow$ Cox PH: $\exp(\hat{\beta})$ = hazard ratio

For the exam

- Be able to **identify** the right model given an outcome description
- Know the key **assumptions** and how to check them:
  - → Proportional odds (ordered logit)
  - → IIA (multinomial logit)
  - → Equidispersion (Poisson)
  - → Proportional hazards (Cox)
- Interpret coefficients correctly:
  - → Ordered/multinomial logit: AMEs or predicted probabilities
  - → Poisson/NegBin: $\exp(\hat{\beta})$ = incidence rate ratio
  - → Cox PH: $\exp(\hat{\beta})$ = hazard ratio
- Know the relevant **R functions**

Questions?